

Eeshaan Jain

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EDUCATION

École Polytechnique Fédérale de Lausanne

3rd year PhD in Computer Science @ *AI in Molecular Medicine Lab* (GPA: **6/6**)

Advisor: *Prof. Charlotte Bunne*

Semester Exchange in Computer Science

Lausanne, Switzerland

Sep '24 – Ongoing

Sep '22 – Feb '23

Indian Institute of Technology Bombay

Dual Degree: B.Tech in Electrical Engineering and M.Tech in Artificial Intelligence (GPA: **9.73/10.0**)

Advisors: *Prof. Abir De*, *Prof. Soumen Chakrabarti*

Institute **Silver Medalist**, Department **Rank 1**

Mumbai, India

Jul '19 – Jun '24

RESEARCH INTERESTS

Generative Modeling, Foundation Models, Agentic Systems, Active Learning, ML Optimization

PUBLICATIONS

- [1] **Eeshaan Jain**, Linus Bleistein, Bart Deplancke, Charlotte Bunne. *Measure Less, Know More: Self-Supervised Test-Time Feature Acquisition*. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] Benedikt von Querfurth*, **Eeshaan Jain***, [...], Olivier Michielin, Charlotte Bunne. *spora: A Unified Multimodal Dataset for Spatial Proteomics*. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
- [3] Johann Wenckstern*, **Eeshaan Jain***, Benedikt von Querfurth, [...], Gabriele Gut & Charlotte Bunne. *The Virtual Tissues foundation model resolves spatial proteomics across scales*. *Nature*, 2026. 
◇ Best Paper Award at *Machine Learning for Genomics Explorations Workshop*, ICLR 2025.
- [4] **Eeshaan Jain***, Kiril Vasilev*, Alexandre Misrahi*, Phil F Cheng, Petros Liakopoulos, Olivier Michielin, Michael Moor, Charlotte Bunne. *MTBBench: A Multimodal Sequential Clinical Decision-Making Benchmark in Oncology*. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. 
- [5] **Eeshaan Jain**, Johann Wenckstern, Benedikt von Querfurth, Charlotte Bunne. *Test-Time View Selection for Multi-Modal Decision Making*. *In Submission*, 2025.
◇ Oral Presentation at *Machine Learning for Genomics Explorations Workshop*, ICLR 2025.
◇ *Generative and Experimental Perspectives for Biomolecular Design Workshop*, ICLR 2025.
- [6] Indradyumna Roy*, Saswat Meher* **Eeshaan Jain**, Soumen Chakrabarti, Abir De. *Graph Matching Systems Deserve Better Benchmarks*. *International Conference on Machine Learning (ICML)*, 2025.
- [7] **Eeshaan Jain***, Indradyumna Roy*, Soumen Chakrabarti, Abir De. *Clique Number Estimation via Differentiable Functions of Adjacency Matrix Permutations*. *International Conference on Learning Representations (ICLR)*, 2025.
- [8] **Eeshaan Jain**, Indradyumna Roy, Saswat Meher, Soumen Chakrabarti, Abir De. *Graph Edit Distance Evaluation Datasets: Pitfalls and Mitigation*. *Learning on Graphs Conference (LOG)* (Extended Abstract), 2024. 
- [9] **Eeshaan Jain***, Indradyumna Roy*, Saswat Meher, Soumen Chakrabarti, Abir De. *Graph Edit Distance with General Costs Using Neural Set Divergence*. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 
◇ Also presented at *Learning on Graphs Conference (LOG)* (Extended Abstract), 2024.
- [10] **Eeshaan Jain**, Tushar Nandy, Gaurav Aggarwal, Ashish Tendulkar, Rishabh Iyer, Abir De. *Efficient Data Subset Selection to Generalize Training Across Models: Transductive and Inductive Networks*. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 

* denotes equal contribution.

TALKS

- **VirTues** [3], **MAVIS** [5], **ECHO-k** [1] @ GenBio, Palo Alto (2026)
- **MTBBench** [4] @ AstraZeneca, Applied Data Science Group (2026)
- **MAVIS** [5] @ MLGenX Workshop, ICLR 2025, Singapore (2025)

FELLOWSHIPS AND AWARDS

- Recipient of the **SwissAI Fellowship (42000\$)** (2025)
- Outstanding research in first year of PhD by **EPFL (600\$)** (2025)
- Awarded research and travel grant by **Broad Institute of MIT and Harvard (3500\$)** for attending ICLR (2025)
- Awarded the **Best Poster Award** at Easter European ML Summer School in Novi Sad, Serbia (2024)
- Awarded the **Institute Silver Medal** at IIT Bombay for outstanding academic performance (#2 rank) (2024)
- Awarded the **Undergraduate Research Award** for the most outstanding Bachelor's Thesis in Electrical Engineering (2023)
- Awarded **Google Conference Scholarship (3000\$)** and **NeurIPS Scholar Award** for attending NeurIPS (2023)
- Awarded the **Institute Academic Prize** for being the top 2 ranks in Electrical Engineering (2021 & 2022)

INDUSTRY EXPERIENCE

Genentech

Jul '26 – Ongoing

ML Intern @ AI & Biology Translation Department and Aviv Regev's Lab

South San Francisco, USA

- Working on efficient generative modeling and theoretical foundations of generative model for active learning, lab-in-the-loop, and Bayesian optimization.

Google Research (Collaboration)

May '22 – May '23

Bachelor's Thesis-I with [Gaurav Aggarwal](#), [Prof. Abir De](#), [Prof. Rishabh Iyer \(UT Dallas\)](#)

Remote

- Associated papers: [10] (NeurIPS 2023)
- Introduced SUBSELNET: A GNN and attention-based model encoder for efficient approximation of outputs across architectures, with trainable, differentiable selectors that relax the combinatorial optimization objective for subset selection. Demonstrated consistent outperformance in accuracy, selection time, and memory efficiency in AutoML tasks, including Neural Architecture Search and Hyperparameter Optimization.

AWL, Inc. Japan, Sony Japan

May '22 – Jul '22

Core AI Team: Scalable Inference on Low-Power Devices via Hardware Optimization

Japan

- Conducted an extensive survey of frameworks for model compression, quantized training, and inference acceleration. Engineered novel CPU-optimized operations with APACHE TVM, achieving up to $8\times$ reduction in inference latency on low-power devices without degrading output fidelity. Integrated GPU-accelerated TENSORRT and DLA pipelines with seamless CPU fallbacks to enhance batched inference efficiency on Jetson platforms.

RESEARCH EXPERIENCE

Foundation Models, Agentic Systems, and Experimental Design in Biology

Sep '24 – Ongoing

Ongoing PhD Research, Advisor: [Prof. Charlotte Bunne](#)

EPFL

- Associated papers: [3] (Nature 2026), [1] (NeurIPS 2026), [2] (NeurIPS 2026), [4] (NeurIPS 2025), [5] (ICLR 2025)
- Pioneered Virtual Tissues (**VirTues**), a foundation model for spatial proteomics that learns multi-scale, marker-aware representations of cells and tissues, enabling zero-shot biomarker discovery and cross-panel generalization.
- Created **MTBBench**, the first multimodal sequential clinical decision-making benchmark modeling real-world oncology workflows to evaluate agentic reasoning with large multimodal models in integrated clinical contexts.
- Developed **MAVIS**, a RL framework that couples foundation models with active test-time experimental design, dynamically selecting diagnostic modalities and features to optimize accuracy, cost, and uncertainty in precision medicine.

End-to-end Differentiable Approaches for Combinatorial Optimization on Graphs

Jul '23 – Jun '24

Master's Thesis, Advisor: [Prof. Abir De](#), [Prof. Soumen Chakrabarti](#)

IIT Bombay

- Associated papers: [9] (NeurIPS 2024 + LoG 2024), [8] (LoG 2024), [7] (ICLR 2025), [6] (ICML 2025)
- Developed end-to-end differentiable methods for APX-Hard combinatorial optimization on graphs and released novel, unbiased datasets for training reasoning algorithms through distant supervision.

Generalization Bounds and Explainability of Graph Neural Networks

May '23 – Aug '23

Research Assistant, Advisor: [Prof. Vikas Garg](#), [Amauri H. Souza](#)

Aalto University

- Derived tighter Rademacher generalization bounds for subclasses of message-passing networks, extending the analysis to temporal GNNs. Established connections between robustness, generalizability, and faithfulness of explainability.

Fairness Audits of Black Box Neural Networks

Jan '23 – May '23

Bachelor's Thesis-II, Advisor: [Prof. Abir De](#)

IIT Bombay

- Developed a comprehensive framework for auditing fairness in black-box models through near-optimal querying under budget constraints. Designed an RL-driven sampling strategy combining a VAE and Gaussian Process surrogate to efficiently explore the input space. Introduced *Disparity*, a novel fairness metric that unifies group and individual fairness, and formulated a disparity-minimization greedy optimization algorithm to mitigate post-audit manipulation effectively.

TECHNICAL PROJECTS

Post-Hoc Out-of-Distribution Detection

Mar '22 - May '22

Advisor: Prof. Sunita Sarawagi (CS 726: Advanced Machine Learning)

IIT Bombay

- Developed a novel OOD detection scoring function based on a Dirichlet distribution assumption over DNN softmax logits, asymptotically interpreted as an ensemble of two positive scoring functions. Demonstrated consistent outperformance across multiple datasets using FPR95, AUROC, and AUPR metrics, while reducing hyperparameter tuning complexity by showcasing the effectiveness of marginless loss functions.

Efficient Matroid-constraint-based Submodular Maximization

Mar '22 - May '22

Advisor: Prof. Ganesh Ramakrishnan (CS 769: Optimization for Machine Learning)

IIT Bombay

- Extended SUBMODLIB by implementing Continuous-Greedy and Accelerated Continuous-Greedy algorithms for maximizing submodular functions under matroid constraints with $(1 - 1/e - \epsilon)$ guarantees. Further, enhanced solution translation through a modified Pipage-Rounding subroutine for efficient conversion of fractional solutions to discrete subsets, and incorporated formulations for the Submodular Welfare Problem, as well as Separable and Generalized Assignment Problems.

SUMMER SCHOOLS

Eastern European Machine Learning Summer School (Google Deepmind)

Novi Sad, Serbia (Jul'24)

Best Poster Award: Efficient Data Subset Selection to Generalize Training Across Models: Transductive and Inductive Networks

Machine Learning for Drug Discovery Summer School (Recursion Pharma)

Montreal, Canada (Jun'24)

Hackathon Winner: Kinase Selectivity Challenge - Predict interactions between proteins and molecules

SCHOLASTIC ACHIEVEMENTS

- Department Rank 1 out of 79 students in the Dual Degree Programme, Electrical Engineering (2020 – 2022)
- Achieved All India Rank 120 in JEE Mains (1.3M candidates) and Rank 355 in JEE Advanced (0.25M candidates) (2019)
- Ranked top 300 across India and appeared in the Indian National Chemistry and Astronomy Olympiads (2019)
- Secured an All India Rank of 100 and received the KVPY Fellowship by Department of Science and Technology (2018)

PROFESSIONAL ACTIVITIES, LEADERSHIP, AND SERVICE

Workshop Organizer

Assisted in organization of workshops at conferences

- Learning Meaningful Representations of Life (LMRL): An ICLR 2025 workshop on multi-scale and multi-modal representation learning from biological data

Student Mentor

Student Mentorship Program: Selected based on a rigorous process of interviews, SOP, and peer reviews

- Institute Student Mentor: Responsible for guiding 12 freshmen through their first year in the institute
- Department Academic Mentor: Mentoring a group of 6 junior undergraduates with their academics and research

Teaching Assistantships | IIT Bombay

Facilitating smooth course organization, grading papers, mentoring students, conducting tutorials and help sessions

- Computer Science: Optimization for Machine Learning, Learning with Graphs, Machine Learning, Intro to CS
- Math and Sciences: Partial Differential Equations, Ordinary, Differential Equations, Quantum Chemistry

Teaching Assistantships | EPFL

Facilitating smooth course organization, grading papers, mentoring students, conducting tutorials and help sessions

- Computer Science: Introduction to Computing, Foundation Models & Generative AI

EXTRA CURRICULAR ACTIVITIES AND OTHER ACHIEVEMENTS

Achievements	- First place (~ 1000\$), Re:AGENT Hackathon in San Fransisco, co-hosted by Anthropic - First place, winning PC setup (~ 6000\$) at Jump Trading Speed Math competition, ICLR 2025 - First place, winning 10000\$ GPU Credits at Jump Trading Speed Math competition, NeurIPS 2024 - Completed all six levels of Speed Arithmetic under IPA, and 2nd in their national-level competition - Winner of the 2020 Poker tournament hosted at IIT Bombay
Volunteering	- Acted as the Lead Convener of the Chemistry Club at IIT Bombay promoting the use of AI in chemistry - Took Python sessions for 1000+ undergraduates and graduates at IIT Bombay over the summer
Others	- Co-founded AISRG – the first student reading group on Artificial Intelligence at IIT Bombay - Elected Class Representative for the Electrical Engineering department for three consecutive years - Successfully completed an year-long training in Chess under National Sports Organization